ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jun 2022 P2HR Worked Solutions

Jun 2022 | P2HR

31 questions ▪ question image followed by worked solution

PREPARED BY

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STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION Q#: 1 Marks: 4

TOPIC: **Transformations**

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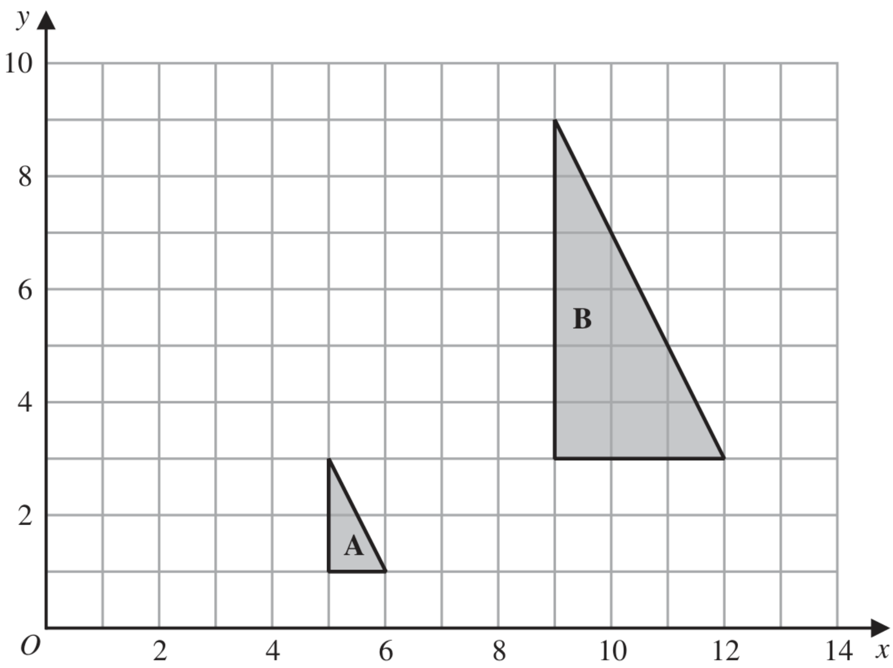
Answer ALL TWENTY FIVE questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

1



(a) Describe fully the single transformation that maps triangle **A** onto triangle **B**

....

....

(3)

(b) On the grid above, translate triangle **A** by the vector $\begin{pmatrix} -4 \\ 3 \end{pmatrix}$

Label your triangle **C**

(1)

(Total for Question 1 is 4 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 4

TOPIC: **Transformations**

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WORKED SOLUTION

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Topic check. Transformations. The tag is correct.

Solution

(a) Triangle A is enlarged to triangle B.
The scale factor is

$$3$$

Using corresponding vertices gives centre $(3, 0)$.
So the transformation is an enlargement, scale factor 3, centre $(3, 0)$.
(b) Translating triangle A by

$$\begin{pmatrix} -4 \\ 3 \end{pmatrix}$$

gives vertices

$$(1, 4),\ (2, 4),\ (1, 6)$$

FINAL ANSWER
(a) enlargement, scale factor 3, centre $(3, 0)$; (b) vertices $(1, 4), (2, 4), (1, 6)$.

PAST PAPER QUESTION

Q#: 2 Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

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2 Write 1200 as a product of powers of its prime factors.
Show your working clearly.

....

(Total for Question 2 is 3 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

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WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Prime factors, HCF and LCM. The tag is correct.

Method

$$1200 = 12 \times 100$$

$$1200 = (2^2 \times 3)(2^2 \times 5^2)$$

$$1200 = 2^4 \times 3 \times 5^2$$

FINAL ANSWER

$2^4 \times 3 \times 5^2$

PAST PAPER QUESTION Q#: 3 Marks: 3

TOPIC: **Statistics Toolkit**

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3 Alberto, Bill, Candela and Diana are four friends.

Here is some information about the height of each of these friends.

Alberto's height is 158 cm.
Bill's height is 175 cm.
Candela's height is greater than Diana's height.

The median height of these four friends is 160 cm.
The range of the heights of these four friends is 21 cm.

Work out Candela's height and Diana's height.

Candela .... cm

Diana .... cm

(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 3

TOPIC: **Statistics Toolkit**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION

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Topic check. Statistics Toolkit. The tag is correct.

Solution

The median is 160. Since the ordered heights are

$$\text{Diana},\ 158,\ \text{Candela},\ 175$$

$$\frac{158 + \text{Candela}}{2} = 160$$

$$\text{Candela} = 162$$

The range is 21:

$$175 - \text{Diana} = 21$$

$$\text{Diana} = 154$$

FINAL ANSWER

Candela = 162 cm, Diana = 154 cm.

PAST PAPER QUESTION Q#: 4 Marks: 5

TOPIC: **Set Notation & Venn Diagrams**

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4 $\mathscr{E}$ = {9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20}
A = {multiples of 3}
B = {odd numbers}

(a) List the members of the set

(i) $A \cap B$

....

(1)

(ii) $A \cup B$

....

(1)

(b) Is it true that $24 \in A$?

Tick one of the boxes below.

Yes ☐ No ☐

Give a reason for your answer.

....

....

(1)

Set C has 4 members such that $C \cap B' = \{10, 18\}$

(c) List the members of one possible set C

....

(2)

(Total for Question 4 is 5 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 5

TOPIC: **Set Notation & Venn Diagrams**

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WORKED SOLUTION

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Topic check. Set notation and Venn diagrams. The tag is correct.

Method

$$\mathcal{E} = \{9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}$$

The multiples of 3 in $\mathcal{E}$ are

$$A = \{9, 12, 15, 18\}$$

The odd numbers in $\mathcal{E}$ are

$$B = \{9, 11, 13, 15, 17, 19\}$$

So

$$A \cap B = \{9, 15\}$$

and

$$A \cup B = \{9, 11, 12, 13, 15, 17, 18, 19\}$$

The statement $24 \in A$ is false because 24 is not in the universal set $\mathcal{E}$.
For C, we need $C \cap B' = \{10, 18\}$ and C has 4 members. One valid choice is

$$C = \{9, 10, 11, 18\}$$

FINAL ANSWER
(a)(i) $\{9, 15\}$, (a)(ii) $\{9, 11, 12, 13, 15, 17, 18, 19\}$, (b) No, (c) one possible set is $\{9, 10, 11, 18\}$

PAST PAPER QUESTION

TOPIC: **Area & Perimeter**

Jun 2022 P2HR - Jun 2022 | P2HR

Q#: 5 Marks: 4

5 The diagram shows a shape made from a square *ABCD* and 4 identical semicircles.

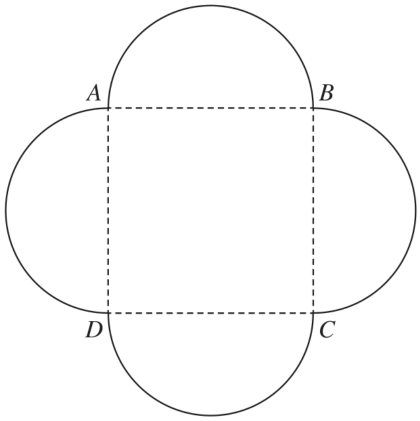

Diagram **NOT** accurately drawn

As shown in the diagram, the semicircles have *AB*, *BC*, *CD* and *DA* as diameters.

The area of the square is 36 cm^2

Calculate the total area of the shape.
Give your answer correct to one decimal place.

.... cm^2

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 4

TOPIC: **Area & Perimeter**

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Topic check. Area & Perimeter. The tag is correct.

Solution

The square has area 36 cm^2, so its side length is

$$\sqrt{36} = 6 \text{ cm}$$

Each semicircle has diameter 6 cm, so radius

$$r = 3 \text{ cm}$$

Area of 4 semicircles:

$$4 \times \frac{1}{2}\pi(3)^2 = 18\pi$$

Total area:

$$36 + 18\pi = 92.548\ldots$$

FINAL ANSWER

92.5 cm^2.

PAST PAPER QUESTION

Q#: 6 Marks: 8

TOPIC: **Solving Linear Equations**

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6 (a) Solve $p = \dfrac{3p - 5}{10}$

Show clear algebraic working.

$p =$ ....
(3)

(b) Simplify a^0 where $a > 0$

....
(1)

(c) Simplify fully $\dfrac{3xy^3}{6x^2y}$

....
(2)

(d) Factorise fully $10c^3d^2 + 15cd^4$

....
(2)

(Total for Question 6 is 8 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 8

TOPIC: **Solving Linear Equations**

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WORKED SOLUTION

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Topic check. Corrected to solving linear equations. The original tag was factorising, but the linear equation is the largest single part.

Solution

(a)

$$p = \frac{3p-5}{10}$$

$$10p = 3p - 5$$

$$7p = -5$$

$$p = -\frac{5}{7}$$

(b) $a^0 = 1$.

(c)

$$\frac{3xy^3}{6x^2y} = \frac{y^2}{2x}$$

(d)

$$10c^3d^2 + 15cd^4 = 5cd^2(2c^2 + 3d^2)$$

FINAL ANSWER

(a) $p = -\frac{5}{7}$, (b) 1, (c) $\frac{y^2}{2x}$, (d) $5cd^2(2c^2 + 3d^2)$

PAST PAPER QUESTION

Q#: 7 Marks: 2

TOPIC: **Algebraic Roots & Indices**

Jun 2022 P2HR - Jun 2022 | P2HR

7 $\dfrac{2^k}{4^n} = 2^x$

Find an expression for x in terms of k and n

$x =$ ....

(Total for Question 7 is 2 marks)

PAST PAPER SOLUTION Q#: 7 Marks: 2

TOPIC: **Algebraic Roots & Indices**

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WORKED SOLUTION

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Topic check. Algebraic Roots and Indices. The tag is correct.

Method

$$\frac{2^k}{4^n} = 2^x$$

Since $4 = 2^2$,

$$4^n = (2^2)^n = 2^{2n}$$

So

$$\frac{2^k}{2^{2n}} = 2^{k-2n}$$

Therefore

$$x = k - 2n$$

FINAL ANSWER

$x = k - 2n$.

PAST PAPER QUESTION Q#: 8 Marks: 3

TOPIC: **Percentages**

Jun 2022 P2HR - Jun 2022 | P2HR

8 A cinema increased the cost of an adult ticket by 12%

After the increase, the cost of an adult ticket was £18.20

Work out the cost of an adult ticket before the increase.

£...

(Total for Question 8 is 3 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 3

TOPIC: **Percentages**

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WORKED SOLUTION

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Topic check. Percentages. The tag is correct.

Method

After a 12% increase, the ticket price is 112% of the original price.

$$18.20 = 1.12 \times \text{original price}$$

$$\text{original price} = \frac{18.20}{1.12} = 16.25$$

FINAL ANSWER

£16.25

PAST PAPER QUESTION

TOPIC: **Powers, Roots & Standard Form**

Jun 2022 P2HR - Jun 2022 | P2HR

Q#: 9 Marks: 4

9 The table gives information about the population, correct to 2 significant figures, of each of five cities in 2018

City	Population (2018)
Ahmedabad	7.7×10^6
Barcelona	5.5×10^6
Chicago	8.8×10^6
Lagos	1.3×10^7
Tokyo	3.7×10^7

(a) Write 8.8×10^6 as an ordinary number.

....

(1)

(b) Which of these cities had the least population in 2018?

....

(1)

(c) Work out the difference between the population of Tokyo and the population of Ahmedabad in 2018
Give your answer in standard form correct to 2 significant figures.

....

(2)

(Total for Question 9 is 4 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 4

TOPIC: **Powers, Roots & Standard Form**

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WORKED SOLUTION

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Topic check. Powers, roots and standard form. The tag is correct.

Method

$$8.8 \times 10^6 = 8800000$$

The least population in the table is Barcelona, with

$$5.5 \times 10^6$$

For the difference between Tokyo and Ahmedabad:

$$3.7 \times 10^7 - 7.7 \times 10^6$$

$$37000000 - 7700000 = 29300000$$

$$29300000 = 2.93 \times 10^7$$

Correct to 2 significant figures:

$$2.9 \times 10^7$$

FINAL ANSWER

(a) 8800000, (b) Barcelona, (c) 2.9×10^7

PAST PAPER QUESTION Q#: 10 Marks: 5

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

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10 The diagram shows triangle *ABP* inside the regular hexagon *ABCDEF*

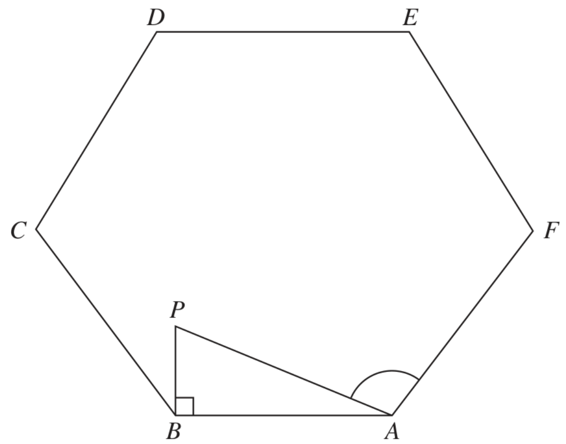

Diagram **NOT** accurately drawn

$AB = 5$ cm $\quad BP = 2$ cm $\quad$ Angle $ABP = 90°$

Work out the size of angle *PAF*
Give your answer correct to 3 significant figures.

.... °

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 5

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

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WORKED SOLUTION

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Topic check. Corrected to Right-Angled Triangles - Pythagoras & Trigonometry. The question needs trigonometry after using the hexagon angle.

Solution

A regular hexagon has interior angle

$$120^\circ$$

In right-angled triangle ABP,

$$\tan(\angle PAB) = \frac{BP}{AB} = \frac{2}{5}$$

$$\angle PAB = \tan^{-1}\left(\frac{2}{5}\right) = 21.801\ldots^\circ$$

Therefore

$$\angle PAF = 120^\circ - 21.801\ldots^\circ = 98.198\ldots^\circ$$

FINAL ANSWER

98.2°.

PAST PAPER QUESTION

Q#: 11 **Marks: 7**

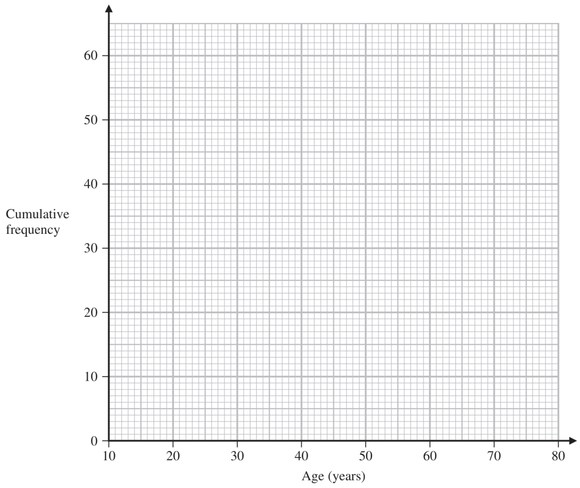
TOPIC: **Cumulative Frequency Diagrams**

Jun 2022 P2HR - Jun 2022 | P2HR

11 The cumulative frequency table shows information about the ages of 60 people who went to a gym on Saturday.

Age (a years)	Cumulative frequency
$10 < a \leqslant 20$	13
$10 < a \leqslant 30$	36
$10 < a \leqslant 40$	42
$10 < a \leqslant 50$	47
$10 < a \leqslant 60$	52
$10 < a \leqslant 70$	56
$10 < a \leqslant 80$	60

(a) On the grid, draw a cumulative frequency graph for the information in the table.



(2)

Question 11 continued

(b) Use your graph to find an estimate for the median of the ages of these people.

.... years

(1)

(c) Use your graph to find an estimate for the interquartile range of the ages of these people.

.... years

(2)

(d) Use your graph to find an estimate for the number of these people who are older than 55 years.

....

(2)

(Total for Question 11 is 7 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 7

TOPIC: **Cumulative Frequency Diagrams**

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WORKED SOLUTION

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Topic check. Cumulative Frequency Diagrams. The tag is correct.

Solution

Plot the points

$$(10, 0), (20, 13), (30, 36), (40, 42), (50, 47), (60, 52), (70, 56), (80, 60)$$

The median is at cumulative frequency 30.
This is between $(20, 13)$ and $(30, 36)$:

$$20 + \frac{30 - 13}{36 - 13} \times 10 = 27.39\ldots$$

Median age is about 27.4 years.
For the interquartile range:

$$Q_1 \text{ is at } 15, \qquad Q_3 \text{ is at } 45$$

$$Q_1 \approx 20 + \frac{15 - 13}{36 - 13} \times 10 = 20.9$$

$$Q_3 \approx 40 + \frac{45 - 42}{47 - 42} \times 10 = 46$$

$$\text{IQR} \approx 46 - 20.9 = 25.1$$

For people older than 55:

$$F(55) \approx 47 + \frac{5}{10}(52 - 47) = 49.5$$

$$60 - 49.5 = 10.5$$

FINAL ANSWER
median about 27.4 years, IQR about 25.1 years, about 11 people older than 55.

PAST PAPER QUESTION

Q#: 12 Marks: 3

TOPIC: **Congruence, Similarity & Geometrical Proof**

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12

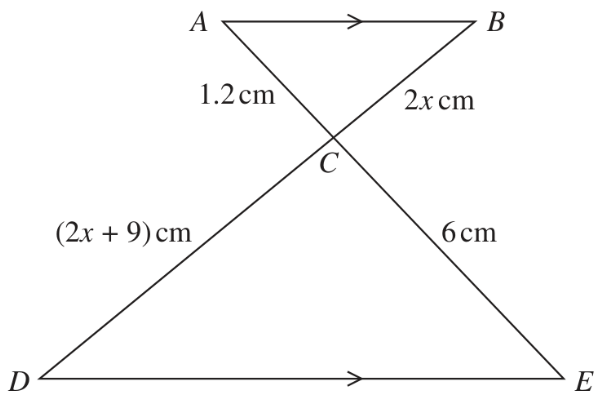

Diagram **NOT** accurately drawn

ACE and *BCD* are straight lines.
AB is parallel to *DE*

Work out the value of x

$x =$ ....

(Total for Question 12 is 3 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 3

TOPIC: **Congruence, Similarity & Geometrical Proof**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION

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Topic check. Corrected to congruence, similarity and geometrical proof. The solution uses similar triangles.

Solution

Since AB is parallel to DE, triangles ACB and ECD are similar.
So corresponding sides are in the same ratio:

$$\frac{AC}{CE} = \frac{BC}{CD}$$

Substitute the lengths:

$$\frac{1.2}{6} = \frac{2x}{2x+9}$$

$$\frac{1}{5} = \frac{2x}{2x+9}$$

$$2x + 9 = 10x$$

$$8x = 9$$

$$x = \frac{9}{8}$$

FINAL ANSWER

$x = \frac{9}{8}$.

PAST PAPER QUESTION **Q#: 13** **Marks: 3**

TOPIC: **Circles, Arcs & Sectors**

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13 The diagram shows a sector *AOB* of a circle with centre *O*

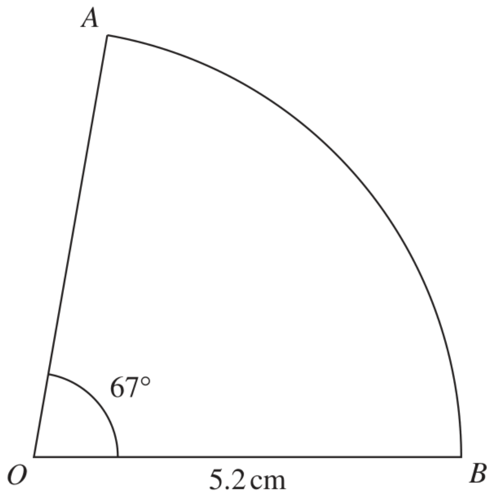

Diagram **NOT** accurately drawn

Angle $AOB = 67°$
$OA = OB = 5.2$ cm

Calculate the perimeter of the sector.
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 3

TOPIC: **Circles, Arcs & Sectors**

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WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Circles, Arcs & Sectors. The tag is correct.

Solution

Arc length:

$$\frac{67}{360} \times 2\pi(5.2) = 6.080\ldots$$

Perimeter:

$$5.2 + 5.2 + 6.080\ldots = 16.480\ldots$$

FINAL ANSWER

16.5 cm.

PAST PAPER QUESTION

Q#: 14 **Marks: 3**

TOPIC: **Combined & Conditional Probability**

Jun 2022 P2HR - Jun 2022 | P2HR

14 Ciara throws **four** fair six-sided dice.

The faces of each dice are labelled with the numbers 1, 2, 3, 4, 5, 6

Work out the probability that at least one of the dice lands on an even number.

....

(Total for Question 14 is 3 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 3

TOPIC: **Combined & Conditional Probability**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Combined & Conditional Probability. The tag is correct.

Solution

The complement of "at least one even number" is "both numbers are odd".

$$P(\text{odd}) = \frac{1}{2}$$

$$P(\text{both odd}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

$$P(\text{at least one even}) = 1 - \frac{1}{4} = \frac{3}{4}$$

FINAL ANSWER

$\frac{3}{4}$.

PAST PAPER QUESTION

Q#: 15 Marks: 3

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jun 2022 P2HR - Jun 2022 | P2HR

15 The diagram shows a kite $ABCD$

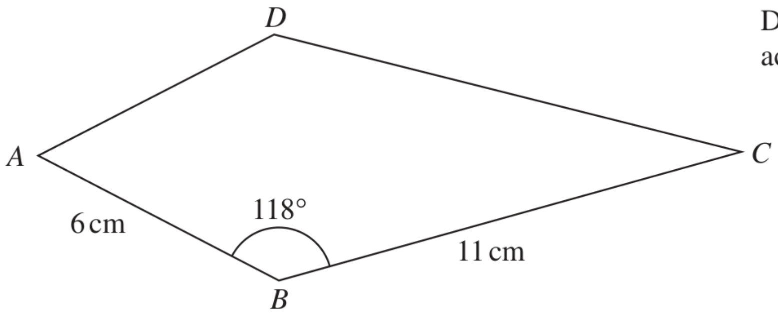


Diagram **NOT** accurately drawn

$AB = 6\,\text{cm}$ $BC = 11\,\text{cm}$ Angle $ABC = 118°$

Calculate the area of the kite.
Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 3

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected to Sine, Cosine Rule & Area of Triangles. The area uses the sine formula.

Solution

A kite can be split into two congruent triangles.
Area of triangle ABC:

$$\frac{1}{2} \times 6 \times 11 \times \sin 118^\circ$$

So the area of the kite is

$$2 \times \frac{1}{2} \times 6 \times 11 \times \sin 118^\circ$$

$$= 6 \times 11 \times \sin 118^\circ = 58.274\ldots$$

FINAL ANSWER
58.3 cm^2.

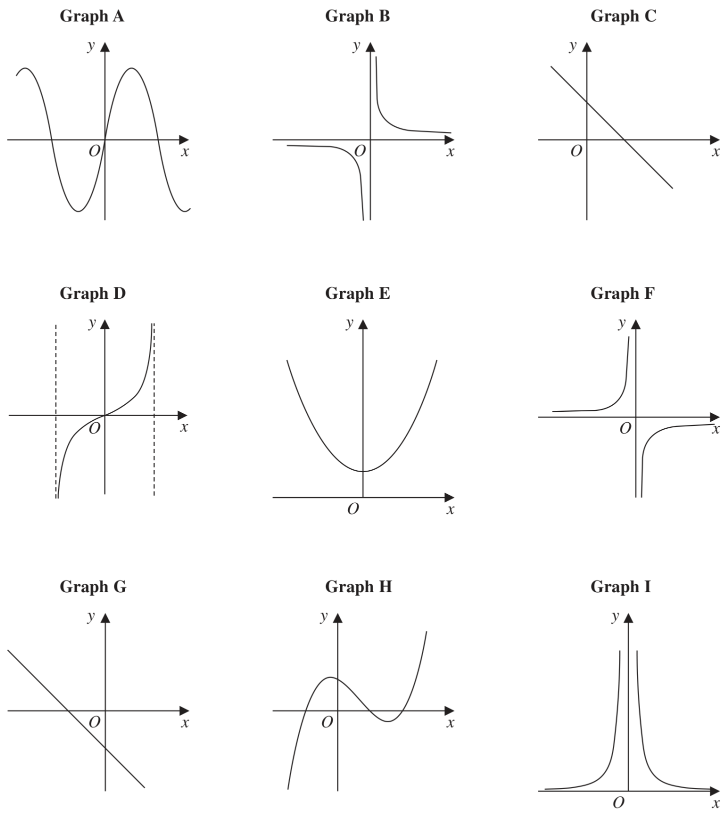
PAST PAPER QUESTION

Q#: 16 Marks: 3

Topic: **Graphs of Functions**

Jun 2022 P2HR - Jun 2022 | P2HR

16 Here are nine graphs.



Complete the table below with the letter of the graph that could represent each given equation.
Write each answer on the dotted line.

Equation	Graph
$y = -2x + 3$	....
$y = -\frac{1}{x}$	....
$y = \tan x°$	....
$y = (x + 1)(x - 1)(x - 2)$	....

(Total for Question 16 is 3 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 3

TOPIC: **Graphs of Functions**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION

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Topic check. Corrected to Graphs of Functions.

Solution

$$y = -2x + 3$$

is a negative straight line with positive y-intercept, so it is **Graph C**.

$$y = \frac{1}{x}$$

is the reciprocal graph in quadrants I and III, so it is **Graph B**.

$$y = \tan x$$

has repeating branches with vertical asymptotes and passes through the origin, so it is **Graph D**.

$$y = (x + 1)(x - 1)(x - 2)$$

is a positive cubic with three roots, so it is **Graph H**.

FINAL ANSWER

C, B, D, H.

PAST PAPER QUESTION

Q#: 17 Marks: 2

TOPIC: **Algebraic Proof**

Jun 2022 P2HR - Jun 2022 | P2HR

17 Use algebra to show that $0.3\dot{4}\dot{5} = \dfrac{19}{55}$

(Total for Question 17 is 2 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 2

TOPIC: **Algebraic Proof**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION

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Topic check. Algebraic proof. The tag is correct.

Solution

Let

$$x = 0.3\dot{4}\dot{5}$$

Then

$$10x = 3.\dot{4}\dot{5}$$

and

$$1000x = 345.\dot{4}\dot{5}$$

Subtract:

$$1000x - 10x = 345.\dot{4}\dot{5} - 3.\dot{4}\dot{5}$$

$$990x = 342$$

$$x = \frac{342}{990} = \frac{19}{55}$$

FINAL ANSWER

$0.3\dot{4}\dot{5} = \dfrac{19}{55}$.

PAST PAPER QUESTION

Q#: 18 Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Jun 2022 P2HR - Jun 2022 | P2HR

18 Kaidan and Sonja went on two different car journeys.

For Kaidan's journey
distance = 80 km correct to the nearest 5 km
time = 2.7 hours correct to 1 decimal place

For Sonja's journey
distance = 33 km correct to 2 significant figures
time = 1 hour correct to the nearest 0.1 hour

Kaidan says,
"My average speed could have been greater than Sonja's average speed."

By considering bounds, show that Kaidan is correct.
Show your working clearly.

(Total for Question 18 is 4 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

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WORKED SOLUTION

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Topic check. Rounding, Estimation and Bounds. The tag is correct.

Method

To show Kaidan's average speed could be greater, compare Kaidan's greatest possible speed with Sonja's smallest possible speed.
For Kaidan:

$$d < 82.5, \quad t \geq 2.65$$

$$\text{greatest speed} = \frac{82.5}{2.65} = 31.13\ldots \text{ km/h}$$

For Sonja:

$$d \geq 32.5, \quad t < 1.05$$

$$\text{smallest speed} = \frac{32.5}{1.05} = 30.95\ldots \text{ km/h}$$

Since

$$31.13\ldots > 30.95\ldots$$

Kaidan's average speed could have been greater.

FINAL ANSWER
Kaidan is correct.

PAST PAPER QUESTION

Q#: 19 Marks: 4

TOPIC: **Functions**

Jun 2022 P2HR - Jun 2022 | P2HR

19 $f(x) = x^2 - 4$

$g(x) = 2x + 1$

Solve $\quad fg(x) > 0$

Show clear algebraic working.

....

(Total for Question 19 is 4 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 4

TOPIC: **Functions**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION

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Topic check. Corrected to Functions. The original tag was solving quadratic equations, but the question is built around composite-function notation.

Solution

$$f(x) = x^2 - 4, \qquad g(x) = 2x + 1$$

$$fg(x) = f(g(x)) = (2x+1)^2 - 4$$

$$= 4x^2 + 4x + 1 - 4 = 4x^2 + 4x - 3$$

Solve

$$4x^2 + 4x - 3 > 0$$

Factorise:

$$(2x+3)(2x-1) > 0$$

The critical values are

$$x = -\frac{3}{2}, \qquad x = \frac{1}{2}$$

The product is positive outside the roots.

FINAL ANSWER

$x < -\frac{3}{2}$ or $x > \frac{1}{2}$.

PAST PAPER QUESTION Q#: 20 Marks: 4

TOPIC: **Coordinate Geometry**

Jun 2022 P2HR - Jun 2022 | P2HR

20 The centre O of a circle has coordinates (4, 7)

The point A, on the circle, has coordinates (6, 11) and AOP is a diameter of the circle.

Find an equation of the tangent to the circle at the point P

....

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 4

TOPIC: **Coordinate Geometry**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The centre of the circle is

$$O = (4, 7)$$

The point A is

$$A = (6, 11)$$

Since AOP is a diameter, O is the midpoint of AP.
Let $P = (x, y)$.

$$\left(\frac{6 + x}{2}, \frac{11 + y}{2}\right) = (4, 7)$$

$$6 + x = 8, \qquad 11 + y = 14$$

$$P = (2, 3)$$

The gradient of OP is

$$\frac{3 - 7}{2 - 4} = 2$$

The tangent is perpendicular to the radius, so its gradient is

$$-\frac{1}{2}$$

Using point $P = (2, 3)$,

$$y - 3 = -\frac{1}{2}(x - 2)$$

$$y = -\frac{1}{2}x + 4$$

FINAL ANSWER

$y = -\frac{1}{2}x + 4$

PAST PAPER QUESTION

Q#: 20 Marks: 4

TOPIC: **Coordinate Geometry**

Jun 2022 P2HR - Jun 2022 | P2HR

20 The centre O of a circle has coordinates (4, 7)

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(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 4

TOPIC: **Coordinate Geometry**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The centre of the circle is

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FINAL ANSWER

$y = -\frac{1}{2}x + 4$

PAST PAPER QUESTION

Q#: 21 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2022 P2HR - Jun 2022 | P2HR

21 Solve the simultaneous equations

$$\begin{aligned} x - 2y &= 3 \\ x^2 - y^2 + 2x &= 10 \end{aligned}$$

Show clear algebraic working.

....

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$x - 2y = 3$$

So

$$x = 2y + 3$$

Substitute into

$$x^2 - y^2 + 2x = 10$$

$$(2y+3)^2 - y^2 + 2(2y+3) = 10$$

$$4y^2 + 12y + 9 - y^2 + 4y + 6 = 10$$

$$3y^2 + 16y + 5 = 0$$

$$(3y+1)(y+5) = 0$$

So

$$y = -\frac{1}{3} \quad \text{or} \quad y = -5$$

Find x:

$$y = -\frac{1}{3} \Rightarrow x = \frac{7}{3}$$

$$y = -5 \Rightarrow x = -7$$

FINAL ANSWER

$(x, y) = \left(\frac{7}{3}, -\frac{1}{3}\right)$ or $(-7, -5)$

PAST PAPER QUESTION

Q#: 21 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2022 P2HR - Jun 2022 | P2HR

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$$x - 2y = 3$$
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Show clear algebraic working.

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(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

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Method

$$x - 2y = 3$$

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Substitute into

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Find x:

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FINAL ANSWER

$(x, y) = \left(\frac{7}{3}, -\frac{1}{3}\right)$ or $(-7, -5)$

PAST PAPER QUESTION

Q#: 22 **Marks: 4**

TOPIC: **Transformations of Graphs**

Jun 2022 P2HR - Jun 2022 | P2HR

22 The point A with coordinates $(-3, 2)$ lies on the straight line with equation $y = f(x)$

(a) Find the coordinates of the image of the point A on the straight line with equation

(i) $y = f(x) - 3$

(.......................... , ...)

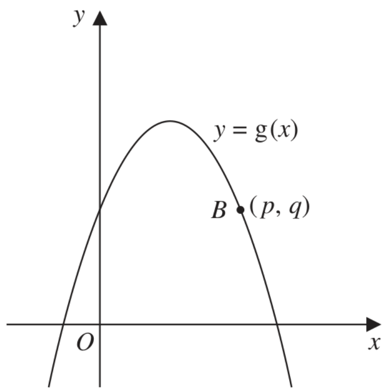
(1)

(ii) $y = f\left(\frac{x}{2}\right)$

(.......................... , ...)

(1)

Here is a sketch of part of the curve with equation $y = g(x)$



The point B with coordinates (p, q) lies on the curve.

(b) Find the coordinates of the image of the point B on the curve with equation

$$y = -g(x - c)$$

where c is a constant.

(.......................... , ...)

(2)

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 4

TOPIC: **Transformations of Graphs**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

Point $A = (-3, 2)$ lies on $y = f(x)$.
For

$$y = f(x) - 3$$

the graph moves 3 units down:

$$(-3, 2) \to (-3, -1)$$

For

$$y = f\left(\frac{x}{2}\right)$$

the x-coordinate is multiplied by 2:

$$(-3, 2) \to (-6, 2)$$

For part (b), point $B = (p, q)$ lies on $y = g(x)$.
For

$$y = -g(x - c)$$

$x - c$ moves the graph c units right, and the minus sign reflects the graph in the x-axis.
So

$$(p, q) \to (p + c, -q)$$

FINAL ANSWER

$(-3, -1)$, $(-6, 2)$, and $(p + c, -q)$

PAST PAPER QUESTION Q#: 22 Marks: 4

TOPIC: **Transformations of Graphs**

Jun 2022 P2HR - Jun 2022 | P2HR

22 The point A with coordinates $(-3, 2)$ lies on the straight line with equation $y = \mathrm{f}(x)$

(a) Find the coordinates of the image of the point A on the straight line with equation

(i) $y = \mathrm{f}(x) - 3$

(.......................... , ...)

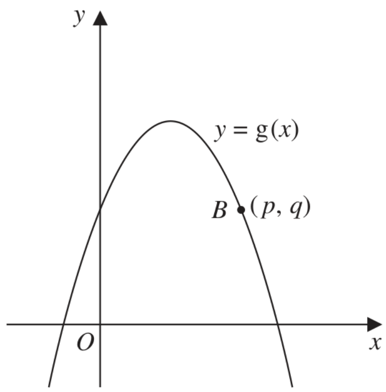
(1)

(ii) $y = \mathrm{f}\left(\dfrac{x}{2}\right)$

(.......................... , ...)

(1)

Here is a sketch of part of the curve with equation $y = \mathrm{g}(x)$



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$$y = -\mathrm{g}(x - c)$$

where c is a constant.

(.......................... , ...)

(2)

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 4

TOPIC: **Transformations of Graphs**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

Point $A = (-3, 2)$ lies on $y = f(x)$.
For

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the graph moves 3 units down:

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For

$$y = -g(x - c)$$

$x - c$ moves the graph c units right, and the minus sign reflects the graph in the x-axis.
So

$$(p, q) \to (p + c, -q)$$

FINAL ANSWER
$(-3, -1)$, $(-6, 2)$, and $(p + c, -q)$

PAST PAPER QUESTION Q#: 23 Marks: 3

TOPIC: **Algebraic Fractions**

Jun 2022 P2HR - Jun 2022 | P2HR

23 Express $\left(\frac{20}{x^2-36}-\frac{2}{x-6}\right)\times\frac{1}{4-x}$ as a single fraction in its simplest form.

....

(Total for Question 23 is 3 marks)

PAST PAPER SOLUTION Q#: 23 Marks: 3

TOPIC: **Algebraic Fractions**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Sine/Cosine Rule to Algebraic Fractions.

Method

$$\left(\frac{20}{x^2-36}-\frac{2}{x-6}\right)\times\frac{1}{4-x}$$

Factor:

$$x^2-36=(x-6)(x+6)$$

So

$$\frac{20}{(x-6)(x+6)}-\frac{2}{x-6}$$

Use the common denominator $(x-6)(x+6)$:

$$\begin{aligned}&\frac{20-2(x+6)}{(x-6)(x+6)}\\&=\frac{20-2x-12}{(x-6)(x+6)}\\&=\frac{8-2x}{(x-6)(x+6)}\\&=\frac{-2(x-4)}{(x-6)(x+6)}\end{aligned}$$

Now multiply by

$$\frac{1}{4-x}=-\frac{1}{x-4}$$

$$\frac{-2(x-4)}{(x-6)(x+6)}\times\left(-\frac{1}{x-4}\right)$$

Cancel $x-4$:

$$\begin{aligned}&\frac{2}{(x-6)(x+6)}\\&=\frac{2}{x^2-36}\end{aligned}$$

FINAL ANSWER

$$\boxed{\frac{2}{x^2-36}}$$

PAST PAPER QUESTION Q#: 23 Marks: 3

TOPIC: **Algebraic Fractions**

Jun 2022 P2HR - Jun 2022 | P2HR

23 Express $\left(\frac{20}{x^2-36}-\frac{2}{x-6}\right)\times\frac{1}{4-x}$ as a single fraction in its simplest form.

....

(Total for Question 23 is 3 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 3

TOPIC: **Algebraic Fractions**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Sine/Cosine Rule to Algebraic Fractions.

Method

$$\left(\frac{20}{x^2-36}-\frac{2}{x-6}\right)\times\frac{1}{4-x}$$

Factor:

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Cancel $x-4$:

$$\begin{aligned}&\frac{2}{(x-6)(x+6)}\\ &=\frac{2}{x^2-36}\end{aligned}$$

FINAL ANSWER

$$\boxed{\frac{2}{x^2-36}}$$

PAST PAPER QUESTION

Q#: 24 Marks: 6

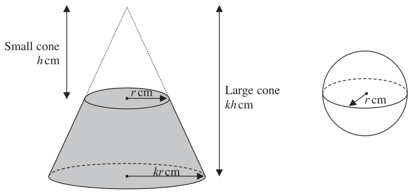
TOPIC: **Area & Volume of Similar Shapes**

Jun 2022 P2HR - Jun 2022 | P2HR

24 The diagram shows a frustum of a cone, and a sphere.

The frustum, shown shaded in the diagram, is made by removing the small cone from the large cone.
The small cone and the large cone are similar.

Diagram **NOT** accurately drawn



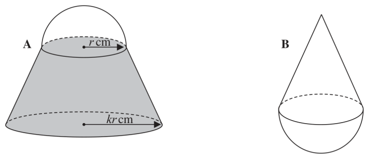
The height of the small cone is h cm and the radius of the base of the small cone is r cm.
The height of the large cone is kh cm and the radius of the base of the large cone is kr cm.
The radius of the sphere is r cm.

The sphere is divided into two hemispheres, each of radius r cm.

Solid **A** is formed by joining one of the hemispheres to the frustum.
The plane face of the hemisphere coincides with the upper plane face of the frustum, as shown in the diagram below.

Solid **B** is formed by joining the other hemisphere to the small cone that was removed from the large cone.
The plane face of the hemisphere coincides with the plane face of the base of the small cone, as shown in the diagram below.

Diagram **NOT** accurately drawn



The volume of solid **A** is 6 times the volume of solid **B**.

Given that $k > \sqrt[3]{7}$

find an expression for h in terms of k and r

h = ...

(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 6

TOPIC: **Area & Volume of Similar Shapes**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

Small cone volume:

$$V_s = \frac{1}{3}\pi r^2 h$$

The larger similar cone has radius kr and height kh, so its volume is

$$V_L = \frac{1}{3}\pi (kr)^2(kh) = \frac{1}{3}\pi k^3 r^2 h$$

So the frustum volume is

$$V_L - V_s = \frac{1}{3}\pi r^2 h(k^3 - 1)$$

The volume of one hemisphere of radius r is

$$\frac{2}{3}\pi r^3$$

Solid A is the frustum plus one hemisphere:

$$V_A = \frac{1}{3}\pi r^2 h(k^3 - 1) + \frac{2}{3}\pi r^3$$

Solid B is the small cone plus one hemisphere:

$$V_B = \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3$$

Given $V_A = 6V_B$:

$$\frac{1}{3}\pi r^2 h(k^3 - 1) + \frac{2}{3}\pi r^3 = 6\left(\frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3\right)$$

Divide by $\frac{1}{3}\pi r^2$:

$$h(k^3 - 1) + 2r = 6h + 12r$$

$$h(k^3 - 7) = 10r$$

$$h = \frac{10r}{k^3 - 7}$$

FINAL ANSWER

$h = \dfrac{10r}{k^3 - 7}$

PAST PAPER QUESTION

Q#: 24 Marks: 6

TOPIC: **Area & Volume of Similar Shapes**

Jun 2022 P2HR - Jun 2022 | P2HR

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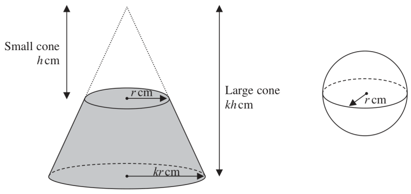

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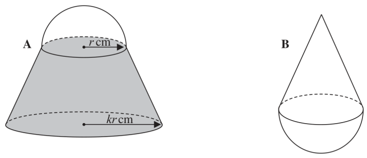

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find an expression for h in terms of k and r

$h =$ ....

(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 6

TOPIC: **Area & Volume of Similar Shapes**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

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So the frustum volume is

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The volume of one hemisphere of radius r is

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Solid A is the frustum plus one hemisphere:

$$V_A = \frac{1}{3}\pi r^2 h(k^3 - 1) + \frac{2}{3}\pi r^3$$

Solid B is the small cone plus one hemisphere:

$$V_B = \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3$$

Given $V_A = 6V_B$:

$$\frac{1}{3}\pi r^2 h(k^3 - 1) + \frac{2}{3}\pi r^3 = 6\left(\frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3\right)$$

Divide by $\frac{1}{3}\pi r^2$:

$$h(k^3 - 1) + 2r = 6h + 12r$$

$$h(k^3 - 7) = 10r$$

$$h = \frac{10r}{k^3 - 7}$$

FINAL ANSWER

$h = \dfrac{10r}{k^3 - 7}$

PAST PAPER QUESTION

Q#: 25 Marks: 5

Topic: **Vectors**

Jun 2022 P2HR - Jun 2022 | P2HR

25 *ABCD* is a parallelogram and *ADM* is a straight line.

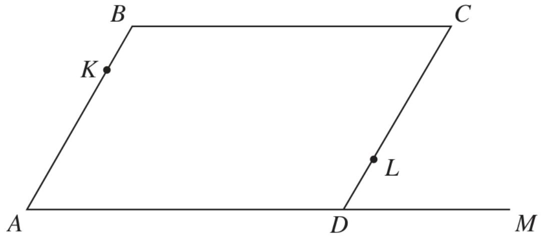

Diagram **NOT** accurately drawn

$\overrightarrow{AB} = \mathbf{a}$ $\quad \overrightarrow{BC} = \mathbf{b}$ $\quad \overrightarrow{DM} = \frac{1}{2}\mathbf{b}$

K is the point on *AB* such that $AK : AB = \lambda : 1$
L is the point on *CD* such that $CL : CD = \mu : 1$
KLM is a straight line.

Given that $\lambda : \mu = 1 : 2$

use a vector method to find the value of λ and the value of μ

$\lambda =$ ....

$\mu =$ ....

(Total for Question 25 is 5 marks)

PAST PAPER SOLUTION Q#: 25 Marks: 5

TOPIC: **Vectors**

Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Linear Graphs to Vectors.

Method

Let A be the origin.

$$\vec{AB} = \mathbf{a}, \qquad \vec{BC} = \mathbf{b}$$

So

$$B = \mathbf{a}, \qquad D = \mathbf{b}, \qquad C = \mathbf{a} + \mathbf{b}$$

Also

$$\vec{DM} = \frac{1}{2}\mathbf{b}$$

So

$$M = \frac{3}{2}\mathbf{b}$$

Point K is on AB:

$$K = \lambda\mathbf{a}$$

Point L is on CD, and $\vec{CD} = -\mathbf{a}$:

$$L = C + \mu(D - C)$$

$$L = \mathbf{a} + \mathbf{b} + \mu(-\mathbf{a})$$

$$L = (1 - \mu)\mathbf{a} + \mathbf{b}$$

Given

$$\lambda : \mu = 1 : 2$$

$$\mu = 2\lambda$$

Since K, L, M are collinear, $\vec{KM}$ is a multiple of $\vec{KL}$.

$$\vec{KM} = -\lambda\mathbf{a} + \frac{3}{2}\mathbf{b}$$

$$\vec{KL} = (1 - \mu - \lambda)\mathbf{a} + \mathbf{b}$$

$$\vec{KL} = (1 - 3\lambda)\mathbf{a} + \mathbf{b}$$

Compare coefficients using

$$\vec{KM} = \frac{3}{2}\vec{KL}$$

$$-\lambda = \frac{3}{2}(1 - 3\lambda)$$

$$-2\lambda = 3 - 9\lambda$$

$$7\lambda = 3$$

$$\lambda = \frac{3}{7}$$

Then

$$\mu = 2\lambda = \frac{6}{7}$$

FINAL ANSWER

$\lambda = \frac{3}{7}, \quad \mu = \frac{6}{7}$

PAST PAPER QUESTION

Q#: 25 Marks: 5

TOPIC: **Vectors**

Jun 2022 P2HR - Jun 2022 | P2HR

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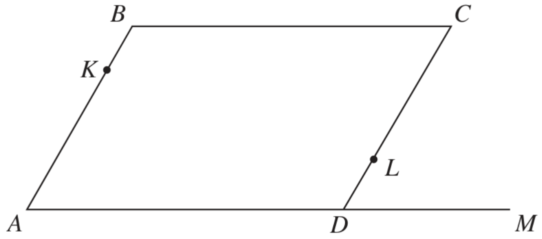

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$\lambda =$ ....

$\mu =$ ....

(Total for Question 25 is 5 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 5

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Jun 2022 P2HR - Jun 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

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So

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Also

$$\vec{DM} = \frac{1}{2}\mathbf{b}$$

So

$$M = \frac{3}{2}\mathbf{b}$$

Point K is on AB:

$$K = \lambda\mathbf{a}$$

Point L is on CD, and $\vec{CD} = -\mathbf{a}$:

$$L = C + \mu(D - C)$$

$$L = \mathbf{a} + \mathbf{b} + \mu(-\mathbf{a})$$

$$L = (1 - \mu)\mathbf{a} + \mathbf{b}$$

Given

$$\lambda : \mu = 1 : 2$$

$$\mu = 2\lambda$$

Since K, L, M are collinear, $\vec{KM}$ is a multiple of $\vec{KL}$.

$$\vec{KM} = -\lambda\mathbf{a} + \frac{3}{2}\mathbf{b}$$

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$$-2\lambda = 3 - 9\lambda$$

$$7\lambda = 3$$

$$\lambda = \frac{3}{7}$$

Then

$$\mu = 2\lambda = \frac{6}{7}$$

FINAL ANSWER

$\lambda = \frac{3}{7}, \quad \mu = \frac{6}{7}$